



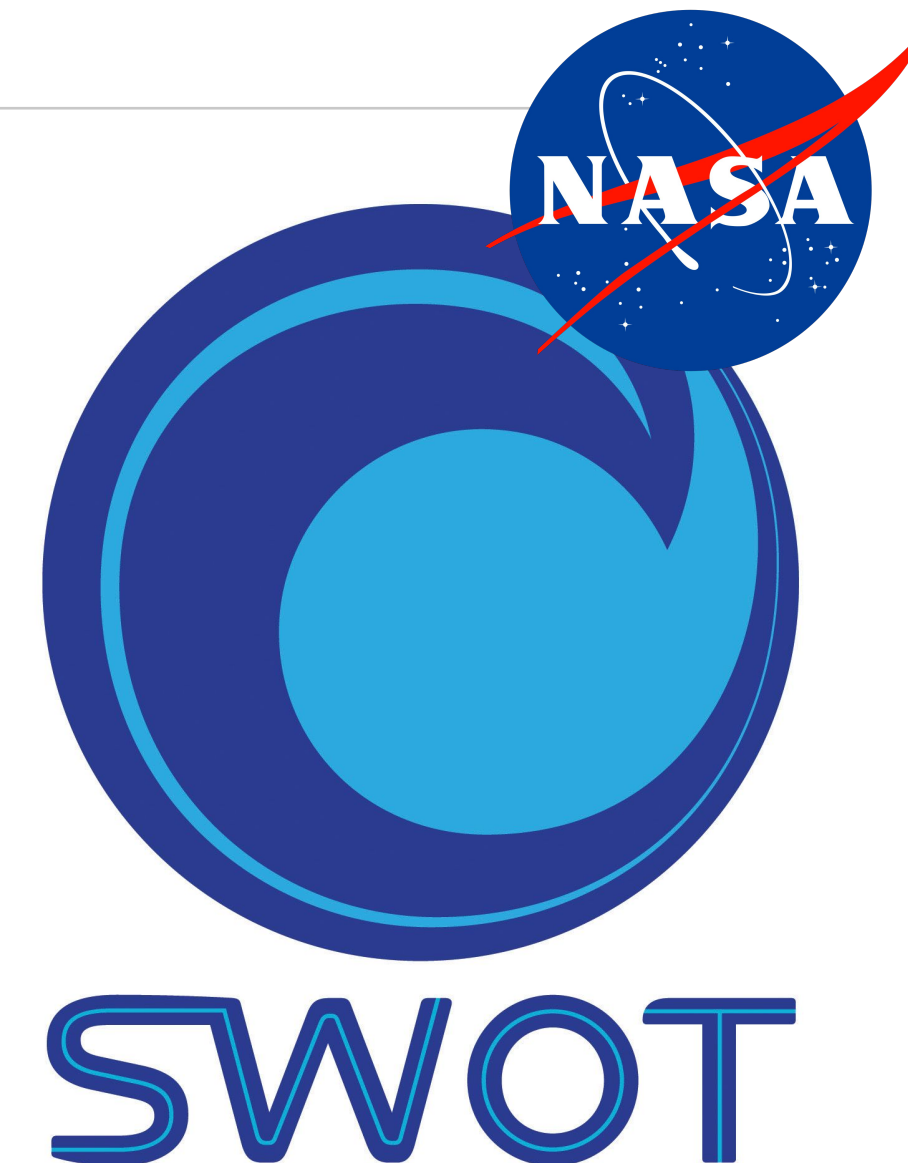
Earth Data Science in the Cloud

Open science in a managed
JupyterHub environment



Tasha Snow, Joanna Millstein, Zachary Katz
SWOT <> CryoCloud Onboarding
Sep 11, 2025

These slides: <https://bit.ly/3VDrfCJ>



Code of Conduct



We strive to create a welcoming and inclusive culture that empowers people to provide outstanding open science. That requires an open exchange of ideas balanced by thoughtful guidelines.

It would be impossible to list everything we can do to create a more welcoming space, and we know this community will find ways to include others in ways that we haven't even thought of. But when in doubt, we encourage you to look to these principles for guidance:

- Practice empathy and humility.
- Participate in an authentic and active way. In doing so, you contribute to the health and longevity of this community.
- Exercise consideration and respect in your speech and actions.
- Assume that everyone we work with is doing their best work.
- Listen carefully and actively.
- Ask questions, and seek to understand your partners' context.
- Encourage other people to listen as much as they speak.
- Attempt collaboration before conflict.
- Treat other people's identities and cultures with respect. Make an effort to say people's names correctly and refer to them by their stated pronouns.
- Be mindful of your surroundings and of your fellow participants. Alert community leaders if you notice a dangerous situation, someone in distress, or violations of this Code of Conduct, even if they seem inconsequential.
- Being respectful of differing opinions, viewpoints, and experiences.
- Giving and gracefully accepting constructive feedback.
- Accepting responsibility and apologizing to those affected by our mistakes and learning from our experiences.
- Focusing on what is best not just for us as individuals, but for the overall community.

Contact Tasha Snow (tsnow03@umd.edu), Joanna Millstein (millstein@mines.edu), or Wilson Sauthoff (sauthoff@mines.edu) to report issues.

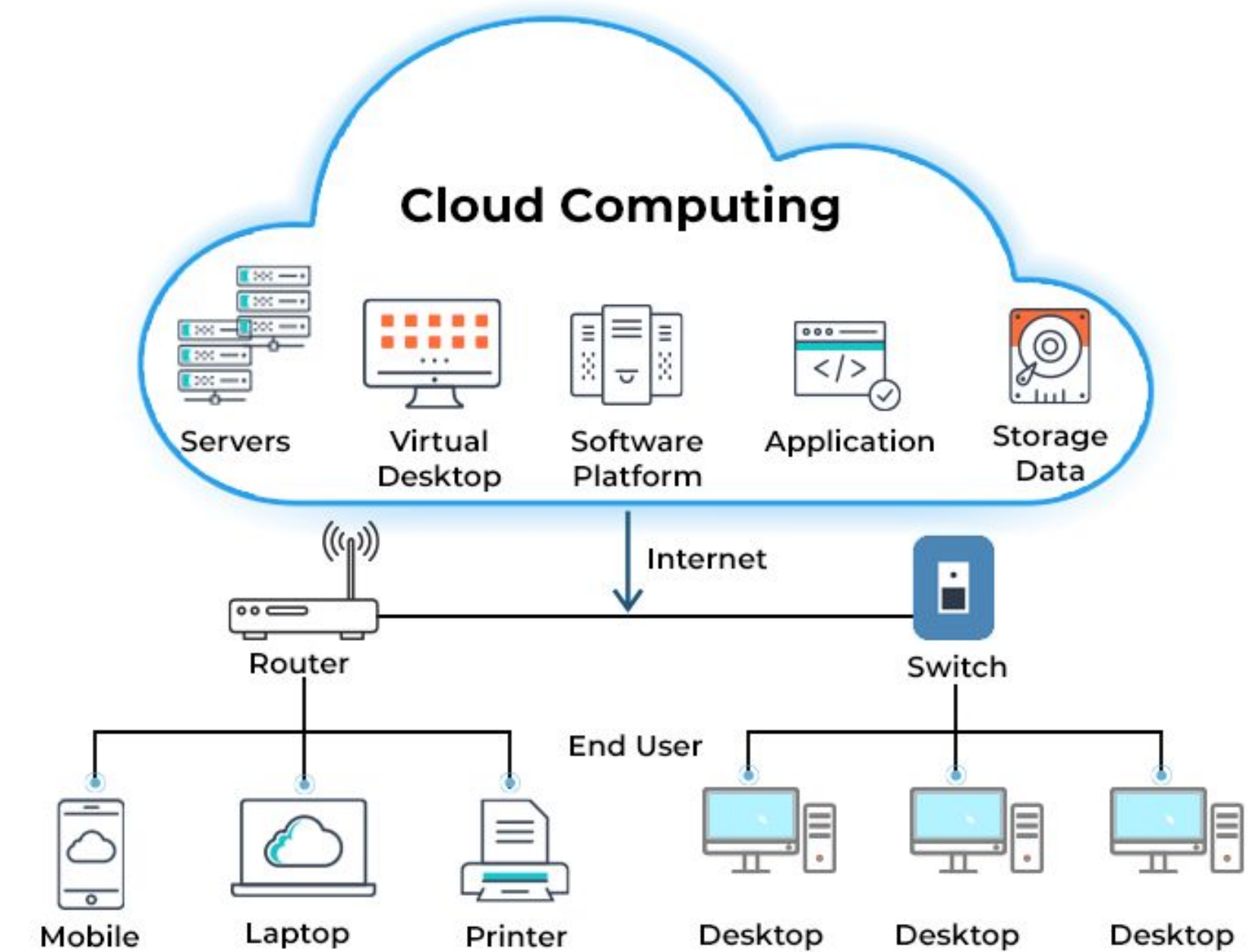
Learning Objectives



- Understand the opportunities and challenges of working in a cloud environment
- Learn about the general technical tools needed for working in the cloud
- Walk through CryoCloud and explore how it will be useful for your research objectives



CLOUD COMPUTING ARCHITECTURE





Ensure you have access to the hub.cryointhecloud.com and fill out the onboarding survey

If you haven't done so, you will also need to fill out the Getting Started Survey to gain access to the hub: <https://forms.gle/TFPzWQkwBPV8nUzY9>



CryoCloud Onboarding Survey:
<https://forms.gle/aFAiXRvH6hqxR2jC6>

What is the Cloud? Data and Analysis on Demand



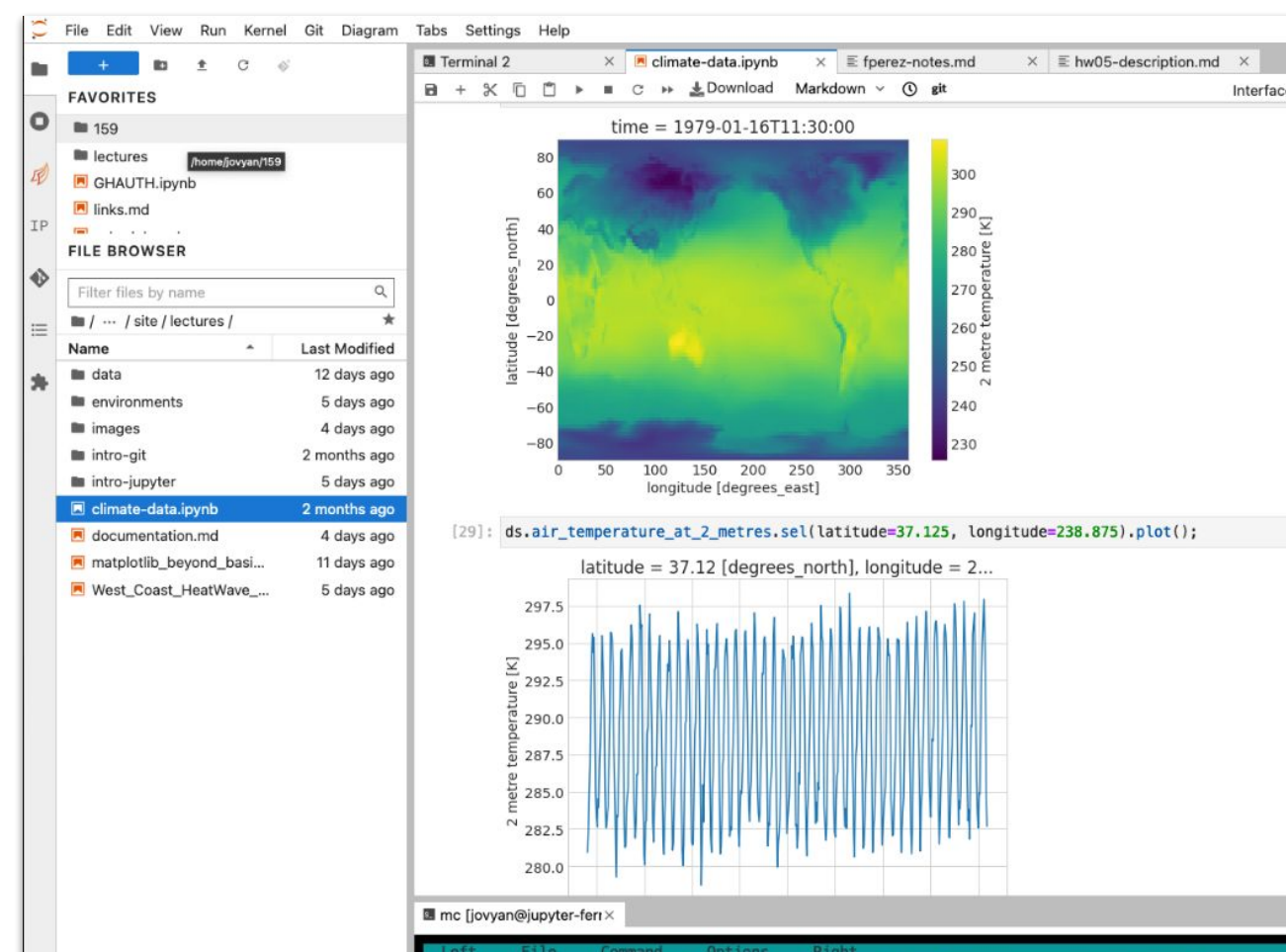
- Instead of downloading data, the cloud lets you bring compute to the data.
 - *[Compute = the processing power (CPUs, GPUs, memory) that runs your programs, models, and analyses]*
- Scalable: from small interactive notebooks to large HPC-like clusters.
- Pay-as-you-go: no need to buy and maintain hardware, but we pay for storage, hours of compute, and egress
- Accessible anywhere with an internet connection.



Barriers to Scaling NASA Earth Data Research



- Data volume is exploding. SWOT alone is streaming an estimated 20+ TB/day.
- Local downloads are inefficient: slow, storage-intensive, and duplicative across users.
- Heterogeneous computing environments lead to reproducibility challenges.
- Users spend more time on data wrangling & infrastructure than science.



hw06-description.md

Homework No 6 - *From Notebooks to Research Packages, Part II*

- Statistics 159/259, Spring 2022
- Due Friday 04/22/2022, 11:59PM PT**
- Prof. F. Pérez and GSI F. Sapienza, Department of Statistics, UC Berkeley.
- This assignment is worth a maximum of **45 points**.
- Assignment type: **individual**.

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Deliverables

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The repository starts with all the code and data with the same layout as LIGO created it. You will reorganize it, while checking that the code still runs, so that all data files live in a `data` directory, generated figures get saved to `figures`, and generated audio goes into `audio`.

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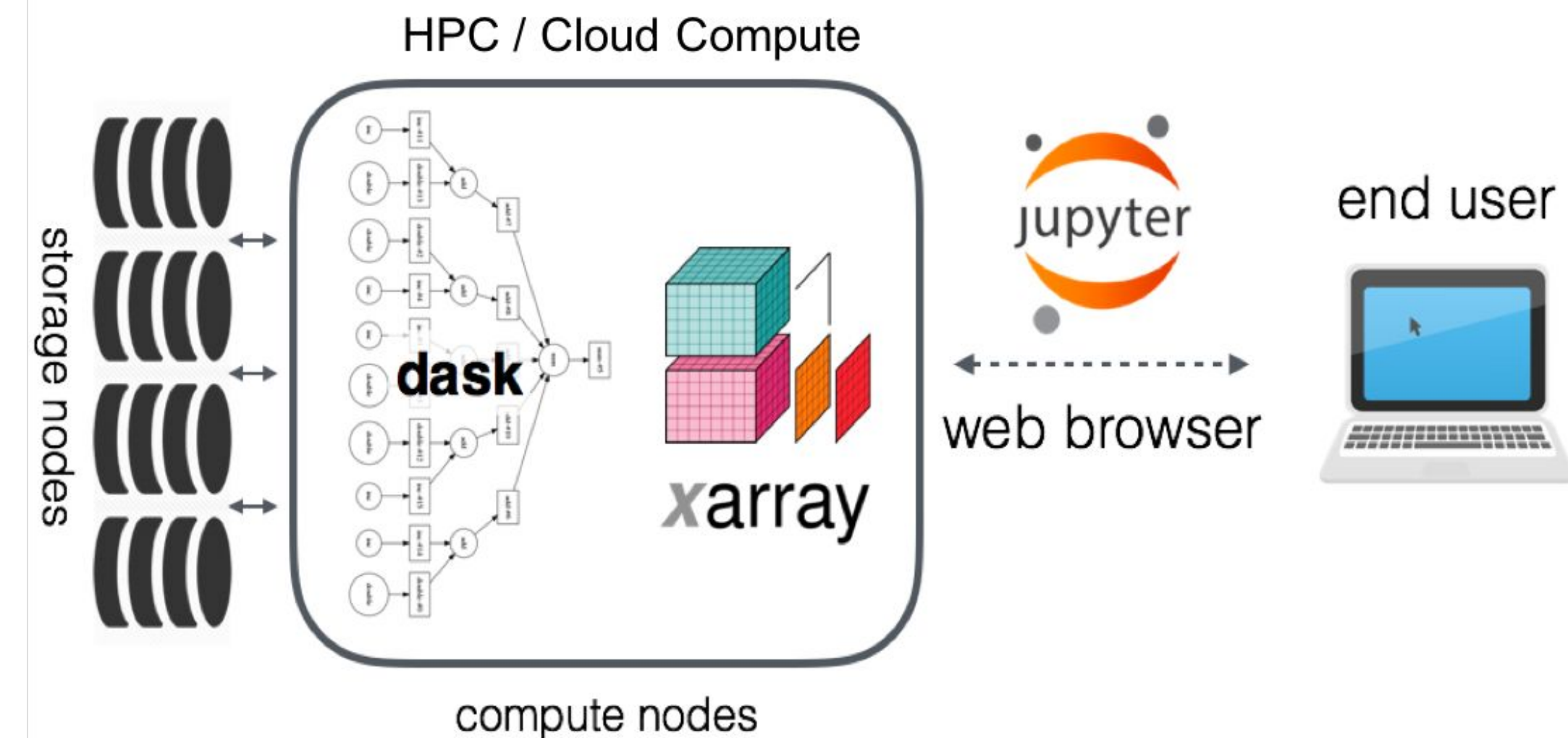
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```



Cloud-Based Research Unlocks New Possibilities



- Co-location of compute and data means that there is no need to move Tbs of data across the internet.
- Elastic scalability: Run anything from a quick notebook to large-scale parallel jobs.
- Collaboration: Shared workspaces, notebooks, and reproducible environments.
- Lower cost of ownership, different cost structure: No local HPC cluster maintenance, pay only for use.
- Reproducibility: Standardized, containerized environments ensure results can be validated.



The Cloud Supports Responsible, Open Science



- FAIR (Findable, Accessible, Interoperable, Reusable): APIs, standards, reusable datasets.
- CARE (Collective Benefit, Authority, Responsibility, Ethics): Respect Indigenous data governance, equitable access.
- Cloud workflows directly advance NASA's open science and equity goals.



To tackle challenges that

- go beyond disciplinary silos...
- require analysis of *really* big data
- integration of disparate data...
- *participation* of disparate, diverse communities...
- to ultimately connect with society and impact critical decision making.

2i2c-Managed Hub Lets us Focus on Science, Not Systems Administration



- Nonprofit [2i2c](#) manages infrastructure, uptime, scaling, and security.
- Researchers avoid system admin overhead (patching, package conflicts, quota issues).
- Benefit from community best practices & continuous interfacing with NASA data users and data engineers.
- Flexible governance: NASA retains oversight while leveraging specialized expertise, secure pipelines, user-friendly analysis portals
- Ensures sustainability beyond individual PI projects or short-term grants.



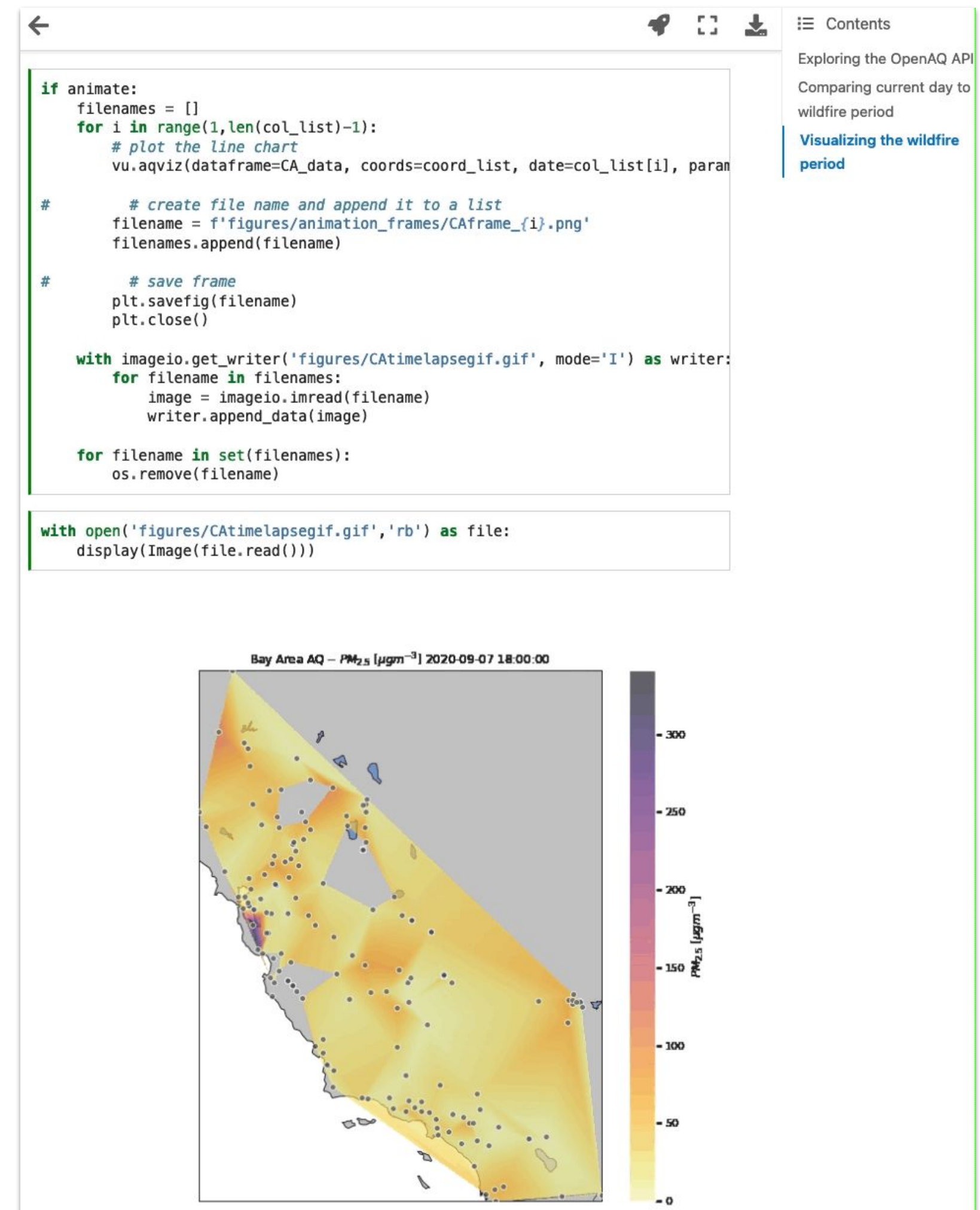
The International
Interactive Computing
Collaboration

2i2c.org

CryoCloud JupyterHub is a Familiar, User-Friendly Gateway



- Browser-based access: no installs, just log in and start coding!
- Integrated with Python, R, Julia, Matlab: all within a comfortable ecosystem for Earth science users.
- Supports scalable backends: single CPU, GPU, or distributed computing via Dask/Spark.
- Consistent environment: centrally maintained kernels and packages.
- Training & onboarding: easier for new users, students, and collaborators.



Rethinking Costs: An Optimized Use of Resources



- Avoid duplicate downloads and redundant local storage across institutions.
- We pay for data storage, hours of compute, and egress [*outbound data transfer from the cloud*]
- Scale up only when needed: this is an efficient use of research budgets.
- Reduce hidden costs of on-prem HPC such as hardware purchases, IT staff, cooling.
- Consolidated cloud infrastructure lowers the total cost of ownership.

A SWOT subset analysis of 10 TB locally requires:

- Expensive hardware
- Hours (days?) to download on a university network.

Same task in cloud:

- Direct query & analysis on data already hosted.
- Compute time ~\$10 (depending on job size).
- **Result: faster, cheaper, no local hardware.**

Building the Next Generation of Earth Scientists



- Browser-based tools lower the entry barrier for Earth science students, no advanced computing skills required.
- Students and early-career scientists train directly in the computing interface of their choice (Matlab, Jupyter Notebooks)
- Skills in Python + cloud computing are portable to academia, industry, and government.
- Enables equitable access: same environment for a student with a laptop as for a senior PI.



Enabling Shared, Scalable Science from SWOT and Beyond

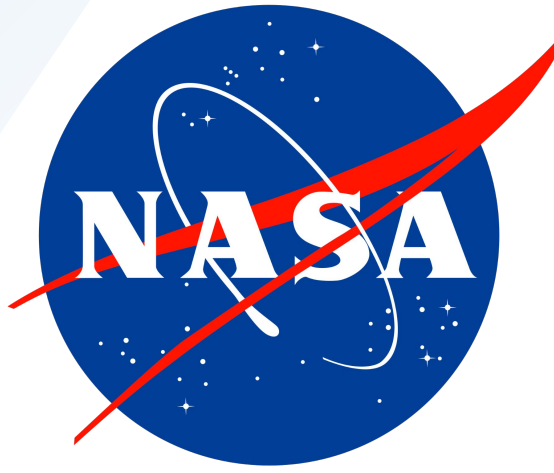


- SWOT: 20+ TB/day of high-res water surface data.
- NISAR: petabyte-scale synthetic aperture radar observations. (80 TB/day)
- Surface Biology & Geology: hyperspectral datasets too large for local workflows.
- *The cloud provides a common infrastructure: same tools, same workflows, different missions.*
- Researchers can work across datasets and perform data fusion for cross-disciplinary work.
- For SWOT:
 - Near-real-time visualization of global water dynamics
 - Large-scale statistical analysis without downloading raw files.
 - Shared workflows for hydrologists, oceanographers, and climate scientists.



**To use the cloud, new community
computing models are needed**

CryoCloud originated to support NASA ICESat-2 Science Team



Standing up usable cloud infrastructure is prohibitively challenging:

- ↳ Non-intuitive pricing structures, documentation, and computing options
- ↳ Costly to use
- ↳ Not obviously more collaborative or faster



ICESAT-2

This didn't ring true to our experience in the cloud!

CryoCloud: A cloud-computing platform with *bumpers*



GOAL: **Simple** and **cost effective** managed cloud environment for **transitioning** and training new users in cloud workflows and **developing** **community best practices**



CryoCloud JupyterBook



CryoCloud

CryoCloud JupyterHub

cryointhecloud.com





CryoCloud JupyterBook



CryoCloud

🔍 Search this book...

Welcome to CryoCloud:
Empowering collaborative Earth
science in the cloud

DETAILS

About
Mission
Getting started
CryoCloud Best Practices
CryoCloud Code of Conduct
Citing CryoCloud

CLOUD TRAINING

Cloud Training

HOW TOS

How To... ✓

TUTORIALS

Tutorials ✓

CONTRIBUTING

Contributing ✓

REFERENCE

Publications, data products, and
libraries supported by CryoCloud

Glossaries

ICESat-2 Resources



☰ Contents

🚀 Our Mission
📖 What You'll Find Here
🌐 Join the Community
💡 Contact Us
Funding Sources

Welcome to CryoCloud: Empowering collaborative Earth science in the cloud

Note

This [JupyterBook](#) is in development as we launch the CryoCloud project.



CryoCloud is a NASA-supported community and cloud platform built in partnership with the International Interactive Computing Collaboration ([2i2c](#)) and designed to advance collaborative, data-intensive Earth science across all disciplines. CryoCloud embraces a broad vision—breaking down silos, fostering intellectual generosity, and empowering scientists to make the most of NASA's vast data resources through streamlined data workflows.

🚀 Our Mission

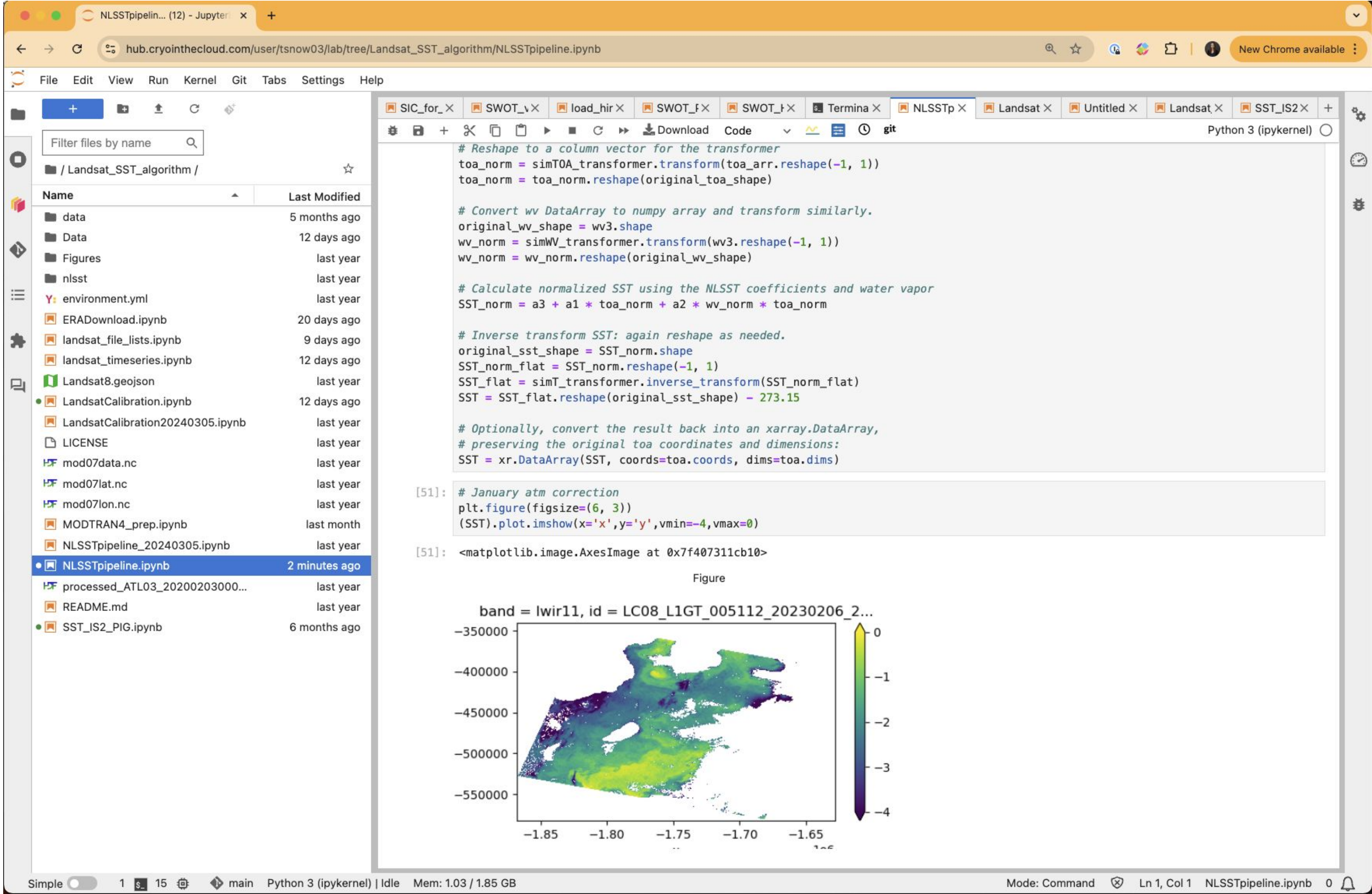
We aim to create an inclusive, interconnected research ecosystem that:

- **Reduces barriers:** Simplifies cloud adoption with cost-efficient and user-friendly tools and data workflows
- **Fosters collaboration:** Brings together data producers, engineers, computational experts, and domain scientists in a shared virtual space
- **Drives innovation:** Accelerates impactful science with cutting-edge workflows and a commitment to open science
- **Promotes sustainability:** Ensures long-term access to state-of-the-art computing and data resources

📖 What You'll Find Here

This JupyterBook is your gateway to the resources, tutorials, and tools developed by and for our community. Whether you're new to cloud computing or an experienced researcher, you'll find:

- Interactive [tutorials](#) to get started with cloud-based data workflows
- Open-source tools and datasets tailored to NASA's Earth Science disciplines



CryoCloud

CryoCloud JupyterHub

A unified experience of research computing: cloud, HPC, local, ...



Remote/local: a unified experience

Hosted Installation



Personal Installation



FAVORITES

159

lectures

GHAUTH.ipynb

links.md

FILE BROWSER

Filter files by name

/ ... / site / lectures /

Name

Last Modified

data

12 days ago

environments

5 days ago

images

4 days ago

intro-git

2 months ago

intro-jupyter

5 days ago

climate-data.ipynb

2 months ago

documentation.md

4 days ago

matplotlib_beyond_basi...

11 days ago

West_Coast_HeatWave_...

5 days ago

Terminal 2

climate-data.ipynb

fperez-notes.md

hw05-description.md

Interface

No Kernel

time = 1979-01-16T11:30:00

latitude [degrees_north]

longitude [degrees_east]

2 metre temperature [K]

ds.air_temperature_at_2_metres.sel(latitude=37.125, longitude=238.875).plot();

latitude = 37.12 [degrees_north], longitude = 2...

2 metre temperature [K]

mc [jovyan@jupyter-fer]

Left

File

Command

Options

Right

~/.n

Size

Modify time

~/.n

Size

Modify time

UP-DIR

1168G/9999G (11%)

Hint: Want your plain shell? Press C-o, and get back to MC with C-o again.

(base) (main) jupyter-fernando-2eperez[hw06]>

1 Help

2 menu

3 view

4 edit

5 copy

6 renMov

7 mkdir

8 delete

9 pullDn

10 quit

hw06-description.md

documentation.md

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JupyterLab: a rich workbench

Markdown w/preview editing (or Python, R, Latex, Bash, C++, ...)

Full terminal workflow

File management

FAVORITES

159

lectures

GHAUTH.ipynb

links.md

FILE BROWSER

Filter files by name

Name

Last Modified

.vnc

159

cloud_science

Desktop

dev

envs

junk

local

make-lesson

scratch

shared

shared-readwrite

site

testing_lesson

tmp

usr

.bash_history

.bash_history-00200.tmp

.bash_history-00478.tmp

.bash_profile

.bash_utils

.bashrc

.condarc

.git-completion.bash

.git-prompt.sh

.gitconfig

.gitconfig.local

.gitignore

.ICEauthority

.lessht

.LICENSE

.python_history

.README.md

.tmux.conf

.wget-hsts

.Xauthority

GHAUTH.ipynb

make-lesson.zip

nbdiff.ipynb

Untitled.ipynb

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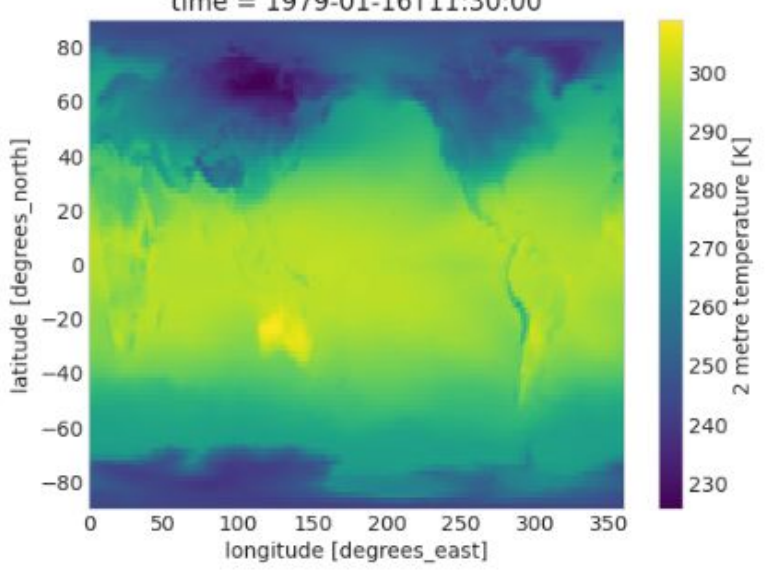
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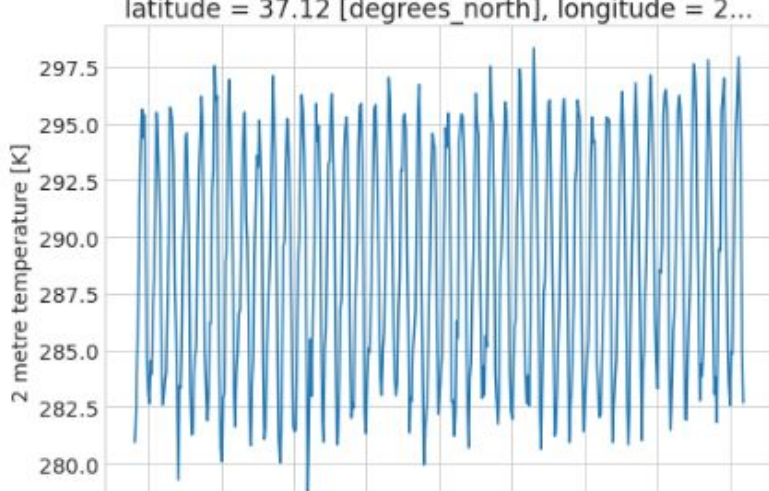
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Left

File

Command

Options

Right

..

NAME

Size

Modify

time

UP—DIR

Apr 6 00:33

README.md

679

Apr 5 23:24

hw05-description.md

6051

Apr 6 00:14

..

NAME

Size

Modify

time

UP—DIR

Apr 6 00:33

ligotools

184

Apr 6 00:37

BBH_events_v3.json

2202

Apr 5 06:26

GW150914_4_template.hdf5

1056864

Apr 5 06:26

H-H1_LOSC_4_V2-1~6259446-32.hdf5

1040592

Apr 5 06:26

L-L1_LOSC_4_V2-1~6259446-32.hdf5

1007420

Apr 5 06:26

README.md

679

Apr 5 23:24

environment.yml

140

Apr 5 06:56

hw06-description.md

804

Apr 6 00:43

index.ipynb

2672144

Apr 6 00:40

UP—DIR

1168G/9999G (11%)

1168G/9999G (11%)

Hint: Want your plain shell? Press C-o, and get back to MC with C-o again.

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1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit

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JupyterLab: a rich workbench

Launcher: not only notebooks, multiple tools under one roof

Data viewing: CSV, images, ...

Applications like a native Diagram editor

The screenshot displays the JupyterLab workbench interface. At the top, a menu bar includes File, Edit, View, Run, Kernel, Git, Diagram, Tabs, Settings, and Help. Below the menu, a toolbar contains icons for creating new files, opening recent files, saving, and other actions. The main workspace is divided into several panes:

- Left Pane:** Contains a 'FAVORITES' section with a list of folders and files, and a 'FILE BROWSER' section showing a tree view of the file system.
- Top Center Pane:** The 'Launcher' view, which displays a grid of application icons. A blue circle highlights this section. The icons include Python 3 (ipykernel), desktop, IPython - mytestenv, LIGO, and syncthing. Below these are icons for a Console, more Python environments, and LIGO. At the bottom of the launcher are icons for Terminal, Diagram, Text File, Markdown File, Python File, and Show Contextual.
- Bottom Left Pane:** A diagram editor interface. A blue circle highlights this section. It features a toolbar with various shapes (arrows, UML, BPMN, flowchart, clipart) and a central canvas where a diagram is being constructed. The diagram includes a globe, a laptop, and a smiley face. To the right of the canvas is a 'Style' panel with options for Fill, Line, Perimeter, and Opacity.
- Right Pane:** Displays a data table with columns: year, month, date_index, fraction_date, c02, and data_adjusted_season. A blue circle highlights this section. Below the table, there is an image of a stained glass window, labeled 'stained_glass_barcelon:'. The image is a colorful, abstract design with various patterns and colors.

The status bar at the bottom shows 'Simple' mode, a zoom level of 2, and memory usage: 'Mem: 298.51 / 8192.00 MB'.

"Normal computing"

Notebook



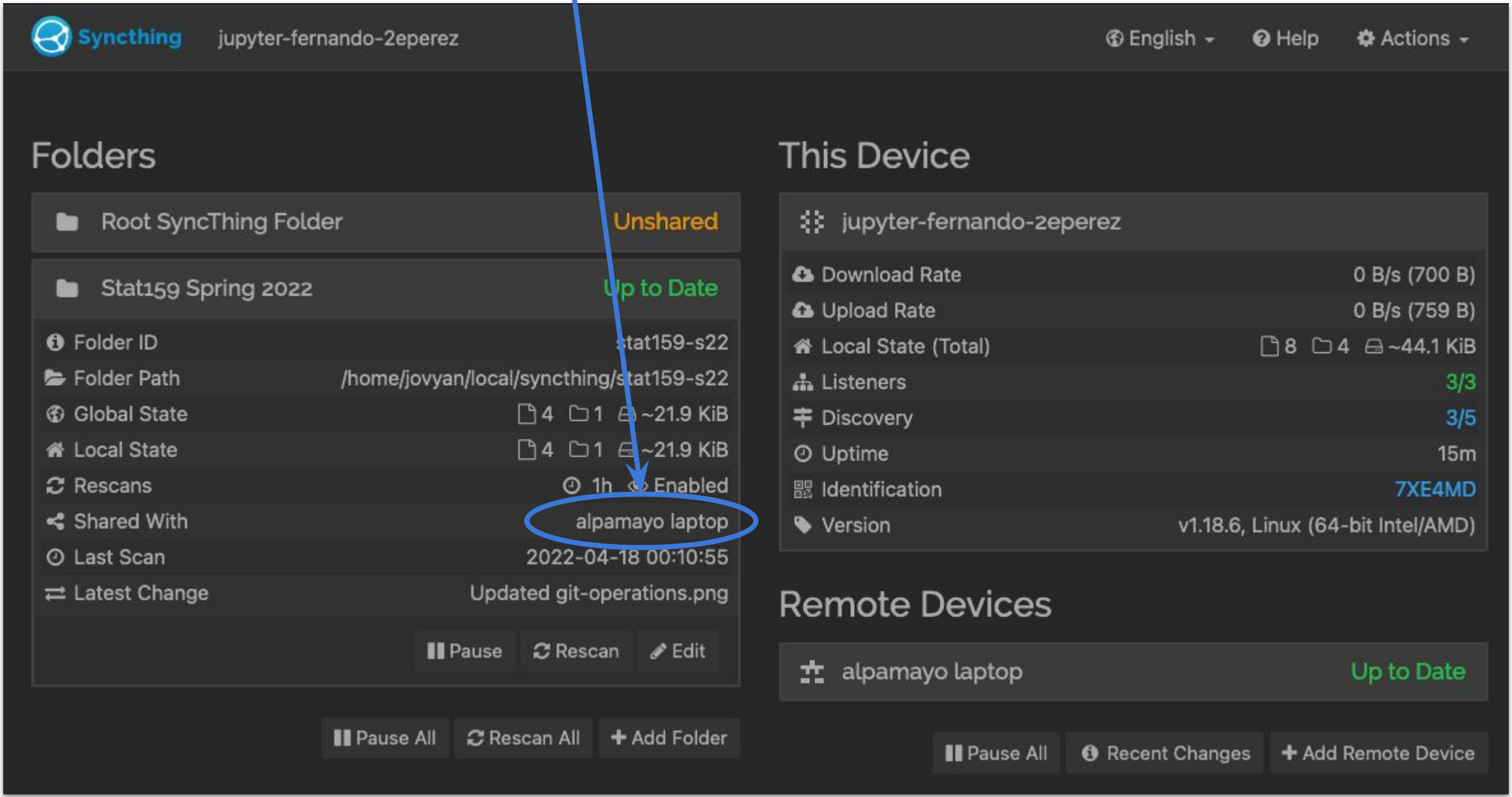
Python 3 (ipykernel)



desktop [↗]

Virtual Linux Desktop

File synchronization with personal machines or other hubs



Versatile choice of computer language and server size

CryoCloud Python Image

Python image maintained by the CryoCloud team

CryoCloud R Image

R environment with many geospatial libraries pre-installed

Matlab

Matlab Environment (bring your own license)

Other...

Build your own image

✓ 1.9 GB RAM, upto 3.7 CPUs

3.7 GB RAM, upto 3.7 CPUs

7.4 GB RAM, upto 3.7 CPUs

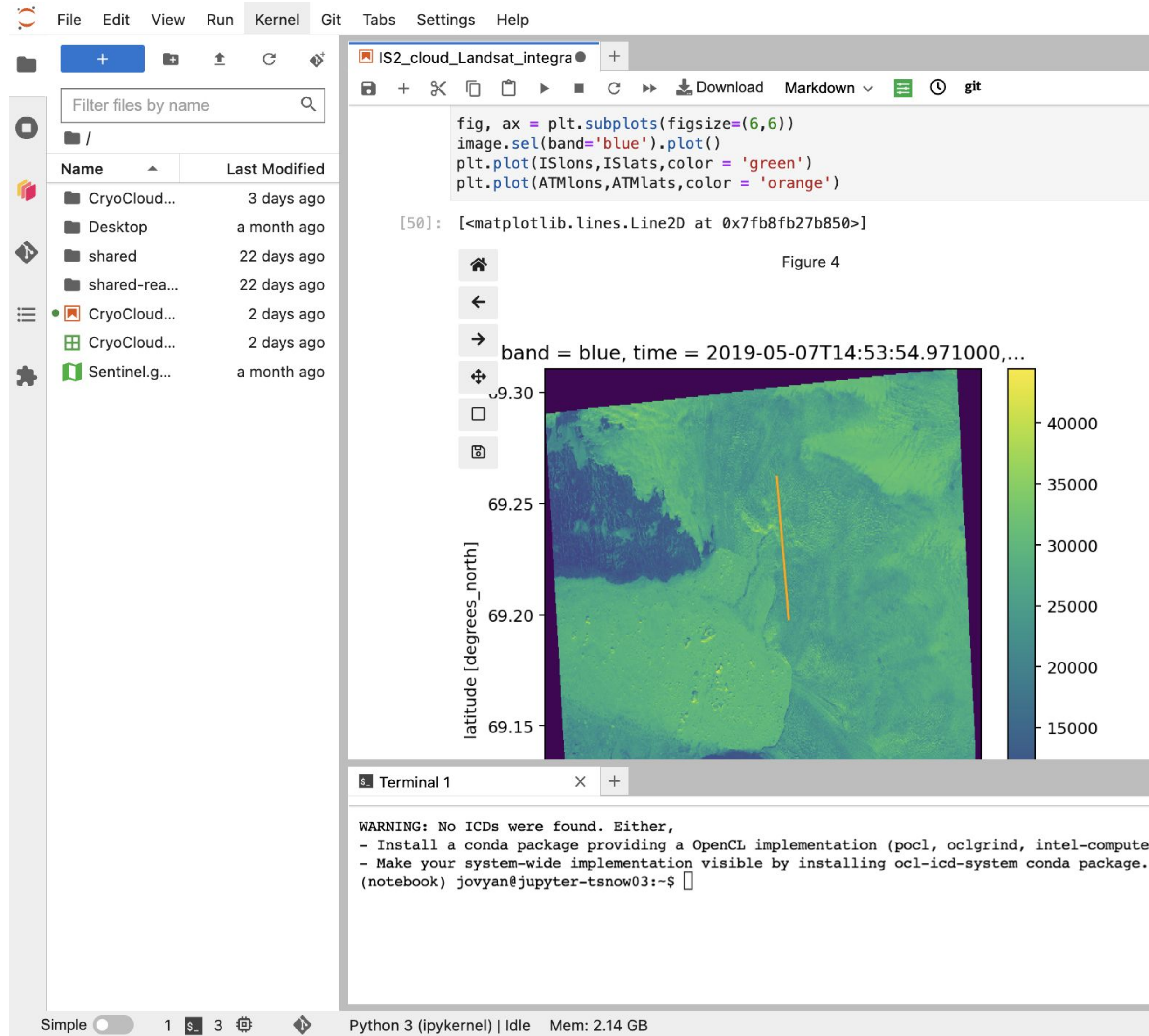
14.8 GB RAM, upto 3.7 CPUs

29.7 GB RAM, upto 3.7 CPUs

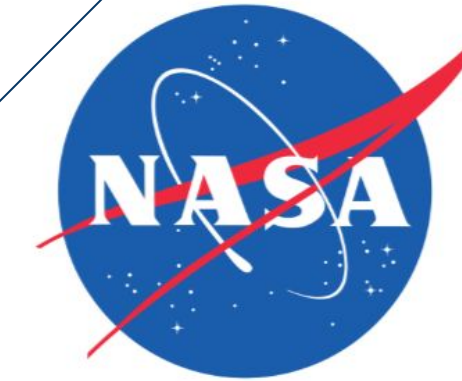
60.6 GB RAM, upto 15.6 CPUs

121.2 GB RAM, upto 15.6 CPUs

Different kinds of users in one place to accelerate feedback and collaboration



Datasets



MODIS
Moderate Resolution Imaging Spectroradiometer



Tools and Developers



earthaccess
A Python Library for NASA Earthdata

Now let's take CryoCloud for a test drive!

hub.cryointhecloud.com



Housekeeping



- ❖ **Personal storage in hub** - keep it to **<10 Gb** unless you talk to us about it
 - \$90 a month to store 2 Tb of data
 - We can work with you to apply for more cloud credits if you need more storage
 - Have a community S3 bucket and temp bucket for larger stores
- ❖ When deleting, make sure to **leave things in `shared-public` that aren't yours**
- ❖ Pip installs on your hub **last only for that session**
 - If multiple people use a package, we will add it to everyone's environment
 - Otherwise you need to install on each use to maintain shareability/collaboration capabilities (**`%pip install package` in your notebook**)
 - Or make own environment in [Python Installation/Environments](#) in the JupyterBook
- ❖ Hub automatically shuts off after 90 min of no use or logging off. Turn off server earlier and save money using **File > Hub Control Panel > push Stop Server**

Troubleshooting



With open science, you have a community of expertise at your fingertips

Open science - we use community platforms and forums for troubleshooting so that others can help provide or learn from our solutions

- **Slack #hub_user_questions** for most questions or issues
- **Github Issues** for JupyterHub or JupyterBook issues:
<https://github.com/CryoInTheCloud/hub-image/issues>
- **Email the CryoCloud team** (cryocloud@mines.edu) if you aren't sure

Citing and contributing to CryoCloud



Please cite CryoCloud in your papers:
<https://doi.org/10.5281/zenodo.7576601>

- Add to the CryoCloud Zenodo Community by adding community-related slides, code, publication pdfs, etc.

CryoCloud links cheatsheet



- Website: cryointhecloud.com
- Hub: hub.cryointhecloud.com
- JupyterBook: book.cryointhecloud.com
- Github: github.com/cryointhecloud
- CryoCloud website code: github.com/cryointhecloud/CryoCloudWebsite
- Slack: cryospherecloud.slack.com
- Pangeo discussions page for troubleshooting: discourse.pangeo.io
- Citing CryoCloud: <https://doi.org/10.5281/zenodo.7576601>
- Zenodo CryoCloud community: <https://zenodo.org/communities/cryointhecloud>

Cool functionality in JupyterHub/Lab



`%` is a magic command that adds functionality

`%matplotlib widget`

Interactive plotting

`%load_ext autoreload`

Autoload imported packages

`%autoreload 2`

`%%time`

Times the cell running

`!` provides access to terminal/command line functions

Shortcuts in Jupyter

Copy Cell(s) notebook:copy-cell	C
Cut Cell(s) notebook:cut-cell	X
Delete Cell(s) notebook:delete-cell	D - D

Insert Cell Above notebook:insert-cell-above	A
Insert Cell Below notebook:insert-cell-below	B
Merge Selected Cells notebook:merge-cells	Shift M

Paste Cell(s) Below notebook:paste-cell-below	V
Redo Cell Operation notebook:redo-cell-action	Shift Z
Run Cell(s) notebook:run-cell	Ctrl Enter
Run Cell(s) and Insert Below notebook:run-cell-and-insert-below	Alt Enter

Efficiently upload large files from your laptop to our S3 bucket

Upload and access (stream, without downloading!) your offset files in our cryocloud aws bucket

```
: # upload

import boto3
import os
import glob
import s3fs
import xarray as xr

s3_client = boto3.client('s3')
```

Upload Data Cube to S3 bucket

```
: fs = s3fs.S3FileSystem()

: file_path = '../Blue_ice_data/offset-ncs/dense_offsets_100m.nc'
  save_path = f"netcdf-sar-data/dense_offsets_100m.nc"

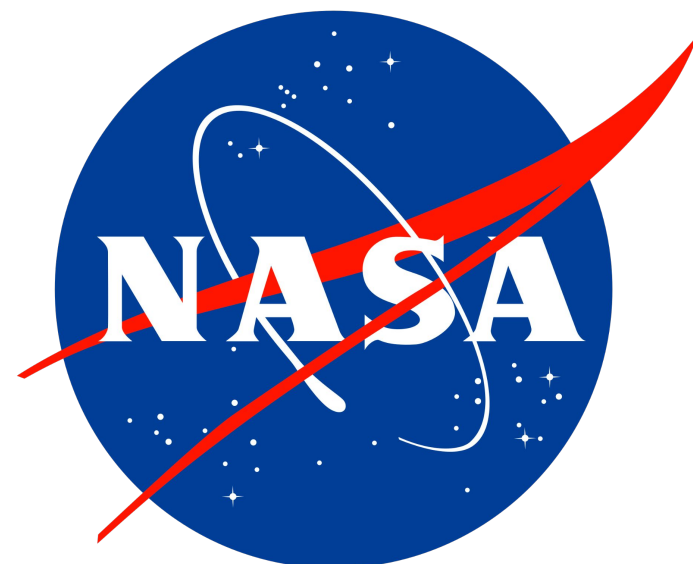
s3_client.upload_file(
    file_path,
    'nasa-cryo-persistent',
    save_path
)
```

Exercise:

Clone the CryoCloud Website into your hub

In a terminal window in CryoCloud:

git clone https://github.com/cryointhecloud/CryoCloudWebsite.git



cryointhecloud.com



CryoCloud

Thank You!

tsnow03.github.io

 @tsnow03

 tsnow03@umd.edu

jdmillstein.github.io

 @jdmillstein

 joanna.millstein@mines.edu